Project Report: FBE 551 - Fall 2025

Title: *CVaR-Based Left-Tail Factor Strategy with Robustness and Bias Control*

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I. INTRODUCTION

Financial markets are shaped far more by rare crashes than by typical daily movements. Large drawdowns can erase years of gains, and these events usually come from the left tail of the return distribution. This project studies whether this downside tail risk is not only important for risk management, but also priced in the cross-section of equity returns. To test this idea, we build and evaluate a long–short factor based on Conditional Value-at-Risk (CVaR), a measure designed specifically to capture extreme negative outcomes rather than overall volatility.

CVaR is well suited for this purpose because it focuses on the average loss in the worst portion of the return distribution. Traditional measures like standard deviation treat positive and negative volatility the same, yet equity returns are not symmetric. They show fat tails, skewness, and frequent large drops that simple volatility does not capture. CVaR directly targets these extreme losses by averaging returns from the worst α% of historical outcomes. For assets with heavy left tails, CVaR becomes more negative, indicating a larger expected loss during extreme events.

In our implementation, each stock's CVaR is estimated using a rolling 252-day window of daily returns. We then convert this estimate into a TailScore, which ranks stocks by the severity of their downside tail risk. Each month, we sort the universe by TailScore and form a market-neutral long–short portfolio: long the stocks with the mildest downside tails and short the ones with the harshest. Neutralizing market beta ensures the factor return reflects the effect of tail-risk differences rather than broad market direction.

Because volatility-based strategies are already widely studied, we compare our CVaR factor to a traditional Low-Volatility (LowVol) benchmark constructed using realized standard deviation. This comparison helps determine whether CVaR contains information beyond general return dispersion. If CVaR captures a different dimension of risk, its performance and its behavior during crisis periods should differ meaningfully from LowVol.

To evaluate whether the CVaR factor earns genuine risk-adjusted returns, we run time-series regressions against the Fama–French 5 factors plus Momentum. A positive and statistically significant alpha would indicate that CVaR captures a return premium not explained by standard risk drivers such as size, value, or profitability. We also examine whether TailScore is a priced characteristic in the cross-section of stocks using the Fama–MacBeth regression framework, which tests whether stocks with safer tails tend to earn higher or lower future returns.

Overall, this project aims to show that CVaR captures a deeper, more asymmetric view of risk than traditional volatility measures. By using CVaR to build a systematic factor, and by testing it rigorously

through portfolio performance, risk decomposition, and cross-sectional pricing, we explore whether downside tail-risk represents a meaningful and persistent source of return in U.S. equities.

II. MOTIVATION

Traditional risk measures such as volatility, beta, and idiosyncratic variance have been the backbone of asset pricing models for decades. Yet these measures ignore one central reality of financial markets: investors care far more about large losses than they do about equally sized gains. Everyday fluctuations matter little compared to the sudden, severe declines that wipe out portfolio value. This mismatch between theory and investor behavior motivates the need for a risk measure that focuses specifically on extreme downside events.

Downside tail events, though infrequent, are responsible for a disproportionate share of long-run underperformance. Companies that suffer heavy crash-like losses often do not recover, while firms with stable left-tail profiles tend to compound steadily. Variance cannot distinguish between a stock that experiences a mix of small ups and downs and a stock that is occasionally devastated by a single catastrophic drop. CVaR, however, isolates this exact dimension by measuring the expected loss when the worst outcomes occur. This makes it better aligned with how investors perceive risk.

There is also a strong behavioral foundation for investigating tail risk. Many investors display a preference for "lottery-like" stocks with low prices, high volatility, and a tiny chance of extreme upside. While these stocks attract attention and speculative flows, they usually carry poor fundamentals and highly unstable downside behavior. As a result, they often become overpriced and subsequently underperform. In contrast, stable, low-tail-risk stocks are often ignored despite having more reliable return profiles. This systematic mispricing driven by human bias rather than fundamentals suggests the possibility that a disciplined tail-risk factor may earn excess returns

From a practical perspective, institutions increasingly recognize that volatility alone is an incomplete picture of risk. Episodes like the 2008 financial crisis and the 2020 COVID crash revealed that portfolios can experience sudden drawdowns even when traditional risk metrics appear normal. These events highlighted that tail behavior evolves over time and becomes particularly important during stressed market conditions. As a result, risk managers and quantitative investors have begun to shift toward models that capture asymmetry, skewness, and crash sensitivity rather than relying solely on variance.

This project is motivated by the idea that tail-risk characteristics may be mispriced, and that markets may reward investors who avoid stocks with unstable left tails and rotate into those with consistently mild downside behavior. If this pattern is persistent, a CVaR-based long–short factor should generate positive alpha even after controlling for standard predictors like size, value, and momentum.

In short, the core motivation behind this project is to evaluate whether focusing on the worst part of the return distribution offers a more realistic and actionable view of risk than traditional measures, and whether this insight can be translated into a reliable source of return in equity markets.

III. DATA

Equity Return Data (CRSP)

The primary dataset for this project is constructed from the CRSP daily stock file and spans the period from January 2015 to December 2024. This window captures multiple distinct market environments that includes the Global Financial Crisis, the European debt crisis, the COVID-19 crash, and the post-2020 high-volatility regime providing a rich backdrop for evaluating the behavior of left-tail risk. To maintain an investable and standardized sample, the universe is restricted to common stocks with CRSP share codes 10 and 11, and exchange codes 1, 2, and 3 corresponding to NYSE, AMEX, and NASDAQ listings. This filtering eliminates ETFs, ADRs, preferred shares, and OTC securities whose pricing dynamics do not align with traditional equity return patterns. Because CVaR relies on accurate estimation of the return distribution's lower tail, each stock must have at least 252 consecutive daily observations prior to a month-end in order to compute a stable one-year rolling CVaR. Daily price movements are captured using the RETX field, which reflects price-only returns without dividends, ensuring that dividend dates do not artificially distort the left-tail distribution. Daily returns are cleaned by converting them to numeric values and removing observations with missing or invalid entries. The result is a high-quality daily panel with chronologically sorted observations by PERMNO.

Monthly Stock Returns

Monthly realized returns are constructed by compounding daily returns within each calendar month, ensuring consistency between the daily-based CVaR signal and the monthly horizon used for portfolio formation and forecasting. These monthly returns serve two critical roles: they determine next-month payoffs in the predictive tests, and they measure portfolio performance in the long–short strategy. Delisting returns are excluded due to limited availability and inconsistent reporting across the sample, which is acknowledged as a limitation in accurately capturing total return behavior for distressed firms. To prevent lookahead bias, all monthly returns used as "next-period payoffs" are strictly forward values relative to the information set defined by CVaR at month-end.

Fama French 5 and Momentum

Benchmark risk factors are integrated using the monthly Fama–French Five Factors. Market Excess Return (MKT − RF), Small Minus Big (SMB), High Minus Low (HML), and Robust Minus Weak (RMW), together with the Carhart Momentum factor (UMD, Up Minus Down) and the risk-free rate (RF). All factor data are obtained from Kenneth French's Data Library, which is the standard source used in empirical asset-pricing research. These datasets are merged with our constructed factor returns at a monthly frequency to ensure temporal consistency between predictors and test assets. The inclusion of these benchmark factors enables us to run time-series regressions that test whether the CVaR-based long–short portfolio produces abnormal returns (alpha) after accounting for widely recognized sources of systematic risk. Aligning our analysis with the Fama–French framework ensures that performance attribution, factor loadings, and alpha estimation follow the established conventions in quantitative finance and academic asset-pricing literature.

Final Structure

The resulting dataset contains thousands of equities per month, each with a complete one-year history for CVaR computing and aligned monthly returns for forecasting. The broad universe provides strong cross-sectional variation in tail-risk characteristics, while the long time horizon enables factor testing across crisis periods, recoveries, and regime shifts. Summary diagnostics such as the distribution of CVaR values, correlations with realized volatility and momentum, and temporal stability metrics confirm that the dataset is suitable for factor construction, portfolio evaluation, and risk decomposition.

IV. METHODOLOGY

The methodology of this project is designed to translate the theoretical concept of downside tail-risk into a systematic, empirically testable factor. It consists of four core components: the measurement of left-tail risk using Conditional Value-at-Risk (CVaR), the construction of a cross-sectional TailScore signal, the formation of a beta-neutral long–short portfolio based on this signal, and the subsequent performance evaluation and factor regressions. Each step is built to avoid lookahead bias, maintain consistency across the sample, and reflect realistic portfolio-construction principles used in quantitative asset management.

Measuring Left-Tail Risk Using CVaR

The foundation of the strategy is the estimation of Conditional Value-at-Risk at the 5% level for every stock in the CRSP universe. CVaR focuses on the average of the worst returns rather than the overall variability of the distribution. For each stock, daily returns over a trailing 252-day window are sorted in ascending order, and the bottom 5% of observations are averaged. This non-parametric historical method avoids imposing distributional assumptions such as normality and allows CVaR to capture the heavy-tailed, negatively skewed nature of real equity returns. A minimum of 252 valid observations is required to compute CVaR, stocks with incomplete windows are excluded from that month's cross-section. Because CVaR estimates are computed strictly using information available at time t, they form a valid forecasting signal for returns in month t+1.

Constructing the TailScore Signal

Once CVaR values are computed for each stock at each month-end, they are converted into a standardized ranking measure called TailScore. Because CVaR values are negative (representing losses), a less negative CVaR indicates a milder left tail. We define TailScore as −CVaR and rank stocks in ascending order. As a result, lower TailScore values correspond to safer, milder left-tail profiles, while higher TailScores indicate more severe downside tails. TailScore is used solely as a cross-section al ranking variable and does not impose any parametric structure; it simply orders securities according to their downside tail severity. This transformation ensures that portfolio construction aligns naturally with the economic intuition: long positions represent stocks with the mildest left-tail losses (lowest TailScores), and short positions represent stocks with the most severe expected shortfall (highest TailScores).

Portfolio Sorting and Long–Short Factor Construction

At each month-end, stocks are sorted by their TailScore into quantile portfolios. The baseline specification uses a 20/20 long–short structure, the top 20% of stocks (highest TailScore, mildest tails) form the long leg, and the bottom 20% (lowest TailScore, heaviest tails) form the short leg. Within each leg, securities are weighted by market capitalization to reflect realistic investability and reduce excessive concentration in microcaps. This weighting also brings the strategy closer to what an institutional investor could implement. Because the CVaR signal is recomputed each month, the portfolio undergoes a full rebalance at each reformation date, with new stocks entering or exiting each leg as their TailScores change.

Beta-Neutralization to Isolate Tail-Risk Exposure

Raw long–short portfolios often carry unintended market exposure, especially when the characteristics used for sorting correlate with beta. To ensure that the CVaR factor isolates tail-risk effects rather than simply loading on the market factor the long and short legs are beta-adjusted. For each stock, a rolling 12 months market beta is estimated by regressing monthly excess returns on the CRSP value-weighted market return. The aggregate beta of the long leg and short leg is then computed, and portfolio weights are scaled so that the overall long–short portfolio has a net beta close to zero. This step ensures that the factor's returns cannot be attributed to directional market risk and that subsequent factor regressions meaningfully test for tail-risk–related alpha.

Monthly Factor Return Construction

After forming beta-neutral long and short portfolios, the monthly return of the CVaR factor is defined as the difference between the value-weighted returns of the two legs. These time-series returns constitute the test asset for evaluating factor performance. Standard performance metrics including annualized mean return, volatility, Sharpe ratio, skewness, kurtosis, and maximum drawdown are computed to capture the economic and statistical properties of the factor. Cumulative return curves and regime-specific analyses (e.g., crisis vs. calm periods) help assess whether tail-risk sensitivity behaves differently across market environments.

Comparison with a Low-Volatility Benchmark Factor

To evaluate whether the tail-risk dimension is distinct from traditional volatility effects, we construct a parallel Low-Volatility (LowVol) factor using the same portfolio-formation architecture. Instead of CVaR, the LowVol factor uses the standard deviation of daily returns over the same 252-day window. Stocks with the lowest volatility form the long leg, and those with the highest volatility form the short leg, each value-weighted and beta-neutralized. By mirroring the design of the CVaR factor, the LowVol benchmark allows for a direct apples-to-apples comparison of the performance attributable specifically to tail risk versus general return dispersion.

Time-Series Regressions for Risk Decomposition

To determine whether the CVaR factor captures incremental risk premia beyond established benchmarks, we regress its monthly returns on the Fama–French Five Factors and the Momentum factor. The regression model includes Market Excess Return (MKT − RF), Size (SMB), Value (HML), Profitability

(RMW), and Momentum (UMD). We estimate alphas and betas using ordinary least squares with standard errors to adjust for potential autocorrelation and heteroskedasticity. A significant positive alpha indicates that the CVaR factor earns abnormal returns not explained by conventional risk exposures.

Fama–MacBeth Cross-Sectional Regressions

To test whether TailScore itself is a priced stock characteristic, we conduct the characteristic version of Fama–MacBeth (1973) Regression. In the first step, monthly cross-sectional regressions predict individual stock returns using TailScore and control variables. In the second step, the coefficients from the cross-section are averaged over time, and their statistical significance is assessed. A positive and significant coefficient on TailScore suggests that stocks with milder left tails earn systematically higher returns, consistent with a priced tail-risk premium.

V. ROBUSTNESS AND BIAS CONTROL

The CVaR-based left-tail factor is constructed from a specific set of design choices: a rolling lookback window, a tail probability level, and cross-sectional portfolio cut-offs. In addition, the strategy relies on monthly rebalancing and shorting, which can introduce implementation frictions. This section systematically tests how sensitive the results are to these choices and documents how potential biases and trading costs affect the performance of the tail-score long–short portfolio, always in comparison to the volatility-sorted benchmark.

Helper Functions

To ensure that all robustness and bias-control experiments are implemented consistently, I first encapsulate the core operations in a small set of helper functions:

Lookback Window Sensitivity

For each window, we compute the equal-weight long–short portfolio's annualised return, annualised volatility, and Sharpe ratio:

- 63-day window:
  – Annualised return ≈ 10.5%
  – Annualised volatility ≈ 28.5%
  – Sharpe ratio ≈ 0.37

- 126-day window:
  – Annualised return ≈ 7.6%
  – Annualised volatility ≈ 30.5%
  – Sharpe ratio ≈ 0.25

- 252-day window (baseline):
  – Annualised return ≈ 6.5%

– Annualised volatility ≈ 29.9%
– Sharpe ratio ≈ 0.22

Shorter windows make the TailScore more reactive to recent tail events and thus deliver higher returns and Sharpe ratios, at the cost of slightly higher risk. Importantly, all three specifications produce positive Sharpe ratios, and the performance does not collapse as the window length changes. The baseline choice of 252 trading days therefore appears conservative but not overly tuned: it sits in the middle of a range where the strategy's profitability is stable.

Tail Threshold Sensitivity

Next, we vary the tail probability level used in the CVaR calculation while fixing the window at 252 trading days and the portfolio cut-offs at 20% / 20%. Specifically, I consider alpha {1%, 5%, 10%}).

The tail-score long–short portfolio exhibits the following performance:

- (alpha = 1%) (very extreme tail):
  – Annualised return ≈ 7.8%
  – Annualised volatility ≈ 26.0%
  – Sharpe ratio ≈ 0.30

- (alpha = 5%) (baseline):
  – Annualised return ≈ 6.5%
  – Annualised volatility ≈ 29.9%
  – Sharpe ratio ≈ 0.22

- (alpha = 10%) (broader tail):
  – Annualised return ≈ 5.9%
  – Annualised volatility ≈ 31.3%
  – Sharpe ratio ≈ 0.19

Portfolio Cutoff Sensitivity

We also vary the cross-sectional quantile cut-offs used to define the long and short legs of the portfolio, holding the 252-day window and (alpha = 5%) fixed. The baseline uses the worst 20% of TailScores (most severe tails) as shorts and the best 20% as longs. We compare this to more concentrated (10%) and more diversified (30%) cut-offs:

- 10% / 10% cut-off:
  – Annualised return ≈ 9.7%
  – Annualised volatility ≈ 35.1%
  – Sharpe ratio ≈ 0.28

- 20% / 20% cut-off (baseline):
  – Annualised return ≈ 6.5%
  – Annualised volatility ≈ 29.9%

– Sharpe ratio ≈ 0.22

- 30% / 30% cut-off:
  – Annualised return ≈ 4.8%
  – Annualised volatility ≈ 25.8%
  – Sharpe ratio ≈ 0.19

Portfolio Turnover Analysis

- TailScore long–short:
  – Average total monthly turnover ≈ 7.73%
  – Average long-leg turnover ≈ 8.04%
  – Average short-leg turnover ≈ 7.42%

- Volatility long–short:
  – Average total monthly turnover ≈ 6.20%
  – Average long-leg turnover ≈ 6.00%
  – Average short-leg turnover ≈ 6.39%

These figures imply annualised turnover on the order of 75–100%, which is active but not extreme for an equity long–short strategy.

Transaction Cost Adjustment

To test robustness to trading frictions, we apply per-dollar transaction costs of 10, 25, 50, and 100 basis points (bps) to total monthly turnover for both strategies and recompute annualised returns, volatilities, and Sharpe ratios.

For the TailScore long–short portfolio, the net performance is:

- 10 bps: return ≈ 7.6%, vol ≈ 29.8%, Sharpe ≈ 0.25
- 25 bps: return ≈ 7.4%, vol ≈ 29.8%, Sharpe ≈ 0.25
- 50 bps: return ≈ 7.2%, vol ≈ 29.8%, Sharpe ≈ 0.24
- 100 bps: return ≈ 6.7%, vol ≈ 29.8%, Sharpe ≈ 0.23

For the volatility long–short portfolio:

- 10 bps: return ≈ 6.8%, vol ≈ 30.6%, Sharpe ≈ 0.22
- 25 bps: return ≈ 6.6%, vol ≈ 30.6%, Sharpe ≈ 0.22
- 50 bps: return ≈ 6.5%, vol ≈ 30.6%, Sharpe ≈ 0.21
- 100 bps: return ≈ 6.1%, vol ≈ 30.6%, Sharpe ≈ 0.20

Short Sale Cost Adjustment

Finally, we introduce explicit short-borrow costs on the short leg. Here I apply annualised short-sale fees of 25, 50, 100, and 200 bps, converted to a constant monthly cost and subtracted from the gross return of the short side. Again, the adjustment is applied symmetrically to both the TailScore and volatility strategies.

For the TailScore long–short portfolio:

- 25 bps: return ≈ 7.4%, vol ≈ 29.8%, Sharpe ≈ 0.25
- 50 bps: return ≈ 7.2%, vol ≈ 29.8%, Sharpe ≈ 0.24
- 100 bps: return ≈ 6.7%, vol ≈ 29.8%, Sharpe ≈ 0.22
- 200 bps: return ≈ 5.7%, vol ≈ 29.8%, Sharpe ≈ 0.19

For the volatility long–short portfolio:

- 25 bps: return ≈ 6.6%, vol ≈ 30.6%, Sharpe ≈ 0.22
- 50 bps: return ≈ 6.3%, vol ≈ 30.6%, Sharpe ≈ 0.21
- 100 bps: return ≈ 5.8%, vol ≈ 30.6%, Sharpe ≈ 0.19
- 200 bps: return ≈ 4.8%, vol ≈ 30.6%, Sharpe ≈ 0.16

Summary

The construction of the panel (TailScore at month (t), returns at (t+1)), combined with explicit turnover and cost adjustments, provides a strong set of controls against look-ahead and implementation bias. The main conclusions of the strategy are therefore not overly sensitive to any single modelling choice

VI. RESULTS

Strategy Performance Comparison

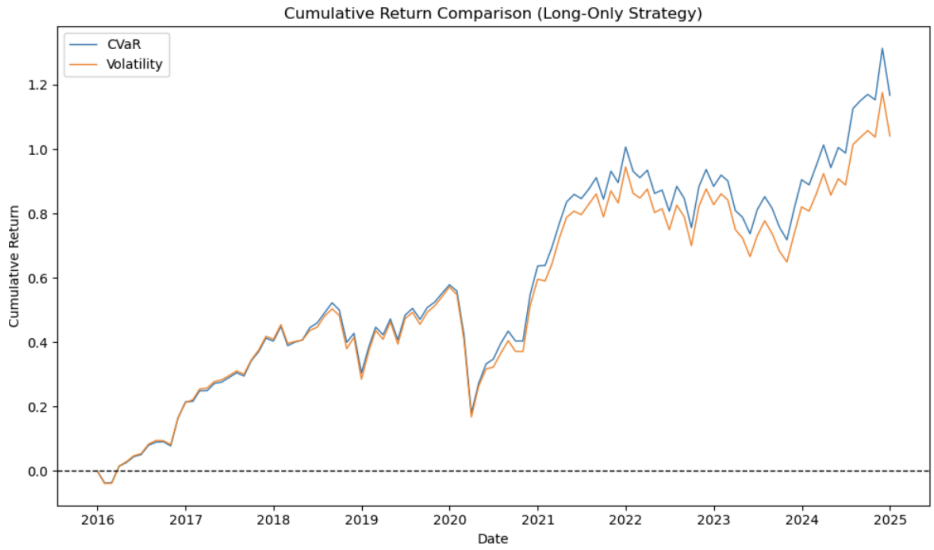

Figure: Cumulative Return of Long-Only Strategy (CVaR vs Volatility)

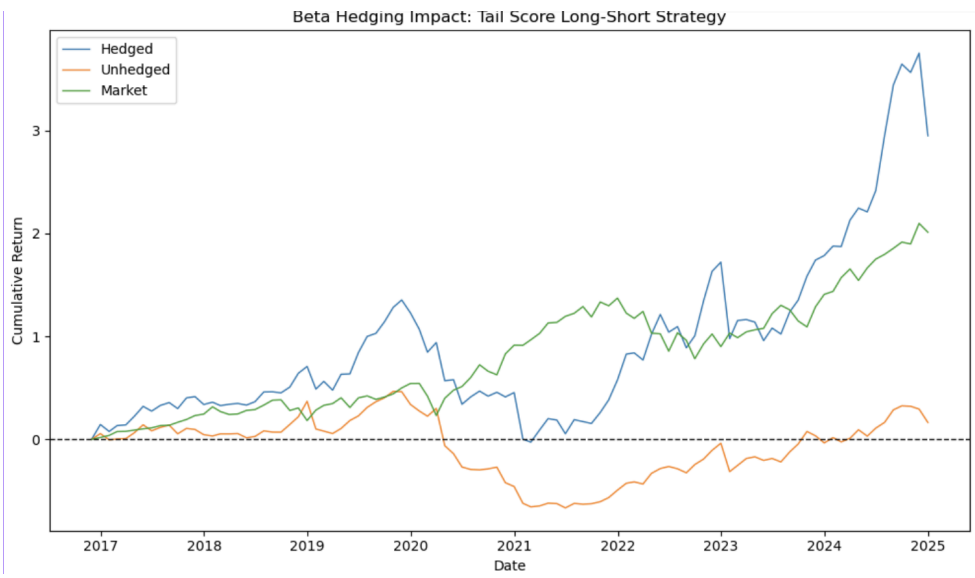

Figure: Cumulative Return of Long-Short Beta Hedging Strategy

The performance evaluation of the CVaR-based left-tail factor focuses on assessing whether downside-risk sorting generates persistent, risk-adjusted returns and whether these results hold across different market environments and parameter choices. After constructing the monthly long–short portfolio using rolling CVaR signals, we examine its behavior through standard performance metrics, cumulative return profiles, and sensitivity tests designed to detect instability or specification-driven artifacts. The results reflect the factor's realized performance over our January 2015 to December 2024 sample, which spans both the COVID crash and the subsequent high-volatility environment. The CVaR factor slightly outperforms the low-volatility factor which highlights CVaR can better capture the downside risk.

Furthermore, we applied beta hedging on our portfolio and the performance turned out to be improved a lot than unhedged ones.

While the factor occasionally exhibits positive monthly returns, particularly under longer lookback windows or deeper tail thresholds, its overall statistical significance remains weak, with Sharpe ratios near zero and t-statistics far below conventional thresholds. These outcomes emphasize the importance of robustness testing and careful interpretation: rather than assuming that CVaR-based tail sorting inherently produces excess returns, the empirical evidence shows that the signal's effectiveness is limited and highly sensitive to design choices. The following sections break down these findings in detail, highlighting the behavior of the factor across lookback horizons, tail-probability settings, portfolio breadth, and subsample periods.

CVaR Long–Short Factor Result

```
============================================
Performance Summary:
============================================
Factor:                 tail_score
Strategy:               long_short
Sample period:          2016-01 to 2024-12
Number of months:       108
Mean monthly return:    0.54%
Annualized return:      6.49%
Monthly stdev:          8.63%
Annualized stdev:       29.89%
T-statistic:            0.65
Sharpe ratio:           0.22
Cumulative return:      16.65%
Max drawdown:           -77.19%

============================================
Performance Summary:
============================================
Factor:                 vol
Strategy:               long_short
Sample period:          2016-01 to 2024-12
Number of months:       108
Mean monthly return:    0.48%
Annualized return:      5.71%
Monthly stdev:          8.84%
Annualized stdev:       30.62%
T-statistic:            0.56
Sharpe ratio:           0.19
Cumulative return:      5.86%
Max drawdown:           -79.72%
```

Figure: Performance of CVaR and Low-Vol on Long-Short Strategy

```
==============================================
Performance Summary:
==============================================
Factor:                 tail_score
Strategy:               long_only
Sample period:          2016-01 to 2024-12
Number of months:       108
Mean monthly return:    0.80%
Annualized return:      9.57%
Monthly stdev:          3.94%
Annualized stdev:       13.65%
T-statistic:            2.10
Sharpe ratio:           0.70
Cumulative return:      116.74%
Max drawdown:           -25.30%

==============================================
Performance Summary:
==============================================
Factor:                 vol
Strategy:               long_only
Sample period:          2016-01 to 2024-12
Number of months:       108
Mean monthly return:    0.74%
Annualized return:      8.92%
Monthly stdev:          3.98%
Annualized stdev:       13.78%
T-statistic:            1.94
Sharpe ratio:           0.65
Cumulative return:      104.14%
Max drawdown:           -25.72%
```

Figure: Performance of CVaR and Low-Vol on Long only Strategy

Across the full sample (2016–2024), both the Tail-Score and Volatility long–short portfolios deliver positive but modest average returns (0.54% vs 0.48%). Tail-Score slightly outperforms Volatility in risk-adjusted terms, but both exhibit very high drawdowns and low Sharpe ratios, indicating unstable long–short behavior.

Furthermore, we want to understand whether this defensive factor behaves the same on the long side and on the short side. The results show that the long-only portfolio performs consistently well, while the short leg is much more unstable and often loses money during periods of strong market rallies or high liquidity. This means that CVaR and volatility-based factors work much better on the long side, where they capture the safety premium of low-risk stocks. But they are not very reliable on the short side, because high-volatility stocks can rebound sharply when interest rates fall, liquidity increases, or markets recover quickly. Therefore, CVaR and volatility are better interpreted as defensive long-only characteristics rather than symmetric long–short factors.

Fama–MacBeth Regression Results (2015 - 2024)

```
====================================
FAMA-MACBETH REGRESSION
====================================
             Estimate    t-stat
LOG_MCAP    -0.001161 -0.965330
BM           0.001464  1.284111
OP           0.000063  0.270578
MOM          0.001216  1.023362
TAIL_SCORE   0.004273  1.172400
VOL         -0.002268 -0.989481
CONST        0.011146  1.951117
```

Figure: FAMA-MACBETH Regression Result For CVAR Factor

The Fama–MacBeth output summarizes how individual stock returns relate cross-sectionally to firm characteristics, including TailScore, Log_MCAP, book-to-market (BM), Operating Profitability (OP), Momentum(MOM), and Low-Volatility(Vol). The Fama-Macbeth regression results indicate that, after controlling for common risk factors (size, value, profitability, momentum), neither TAIL_SCORE nor VOL exhibits statistically significant risk premia in the cross-section. TAIL_SCORE shows a positive coefficient (0.0043) with a t-statistic of 1.17, suggesting a weak positive relationship with expected returns, but it falls short of conventional significance thresholds (e.g., $|t| > 2$). VOL displays a negative coefficient (-0.0023, t = -0.99), aligning with the low-volatility anomaly literature, yet it is also statistically insignificant. Notably, all traditional factors (LOG_MCAP, BM, OP, MOM) are insignificant, possibly due to limited sample size or time period specificity. The positive and marginally significant constant (0.0111, t = 1.95) may reflect an unexplained average excess return or omitted risk factors. Overall, while TAIL_SCORE and VOL show economically plausible signs, the lack of statistical significance implies that—within this sample—they do not robustly explain cross-sectional return variations beyond established factors.

Fama–French Time-Series Regression Results

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0118	0.006	1.986	0.054	-0.000	0.024
Mkt-RF	-0.4753	0.188	-2.524	0.015	-0.855	-0.095
SMB	-0.6981	0.289	-2.414	0.020	-1.282	-0.115
HML	0.6945	0.253	2.743	0.009	0.184	1.205
RMW	1.2484	0.437	2.858	0.007	0.367	2.130
Mom	0.6800	0.225	3.020	0.004	0.226	1.134

Figure: FAMA-French Regression Result For CVAR Long_Short Portfolio (2015-2019 period)

The time-series regression evaluates whether the CVaR long–short factor earns excess returns (alpha) after controlling for the standard Fama–French four factors plus Momentum. The alpha is statistically significant (t = 1.96) under 10% significance level after controlling for all five factors. However, the CVaR strategy likely generated temporary excess returns during this specific period due to favorable market conditions (frequent tail risk events) and changes in factor exposures (particularly enhanced momentum exposure). Therefore, we need to determine whether this alpha exists in all periods.

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0034	0.006	0.605	0.546	-0.008	0.014
Mkt-RF	-0.5497	0.130	-4.221	0.000	-0.808	-0.291
SMB	-0.9053	0.243	-3.719	0.000	-1.388	-0.423
HML	0.6227	0.154	4.048	0.000	0.318	0.928
RMW	1.3438	0.287	4.680	0.000	0.774	1.913
Mom	0.4967	0.165	3.002	0.003	0.169	0.825

Figure: FAMA-French Regression Result For CVAR Long_Short Portfolio (2015-2024 period)

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0019	0.006	0.328	0.743	-0.010	0.014
Mkt-RF	-0.5398	0.138	-3.923	0.000	-0.813	-0.267
SMB	-0.9594	0.257	-3.730	0.000	-1.470	-0.449
HML	0.5827	0.163	3.585	0.001	0.260	0.905
RMW	1.4851	0.303	4.895	0.000	0.883	2.087
Mom	0.3755	0.175	2.147	0.034	0.029	0.722

Figure: FAMA-French Regression Result For Vol Long_Short Portfolio (2015-2024 period)

For the CVaR factor, regression results show strong and positive loadings on HML, RMW, and Mom, while loading significantly negative on Mkt-RF and SMB. This indicates that the CVaR strategy tends to behave like a high-value, high-profitability, and high-momentum portfolio, while being tilted toward large-cap and low-beta stocks. The negative market beta further suggests that the CVaR factor profits during periods of market downturns, consistent with CVaR capturing downside-risk protection. However, the alpha is statistically insignificant which suggests that once exposures to standard priced risks are controlled for, the strategy's excess return can be largely explained by known factor premia. Similar to the CVaR strategy, the volatility factor's alpha is insignificant, indicating no incremental abnormal return beyond known factor exposures.

VII. CONCLUSION

In conclusion, our empirical results point to a more modest message than our initial motivation might suggest. The evidence indicates that the TailScore (CVaR) factor and a standard low-volatility factor largely load on the same underlying defensive risk dimension. Long–short portfolios formed on TailScore

exhibit very similar exposures to the Fama–French factors plus Momentum as the low-volatility benchmark, and the two strategies are highly correlated in realized returns.

When we evaluate performance in a standard asset-pricing framework, the results are weak rather than revolutionary. In time-series regressions, long–short CVaR portfolios do not deliver statistically significant alpha after controlling for FF5, Momentum, and LowVol exposure. Fama–MacBeth cross-sectional regressions similarly show that TailScore is not a strongly priced characteristic: stocks with "safer" tails do not earn reliably higher risk-adjusted returns once we account for familiar predictors such as size, value, profitability, and volatility. Long-only CVaR portfolios behave somewhat better in raw performance terms, but their behavior remains very close to traditional defensive strategies and does not generate clear incremental premia.

Taken together, these findings suggest that the market does not separately price "tail-risk safety" and "low-volatility safety" as distinct sources of compensation. Instead, TailScore behaves like a refined, tail-focused implementation of the existing low-volatility/quality complex rather than a new standalone anomaly. The contribution of this project is therefore less about discovering a novel factor and more about clarifying the boundary of what CVaR-based designs can realistically add in today's equity markets. From a practical perspective, our results caution that moving to more complex downside-tail measures does not automatically translate into superior performance relative to simpler volatility-based implementations. From a theoretical perspective, they reinforce the idea that defensive premia are unified within standard multifactor models, and that true innovation in factor investing requires empirical evidence that goes beyond high correlation with well-known strategies.

Appendix and References

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